

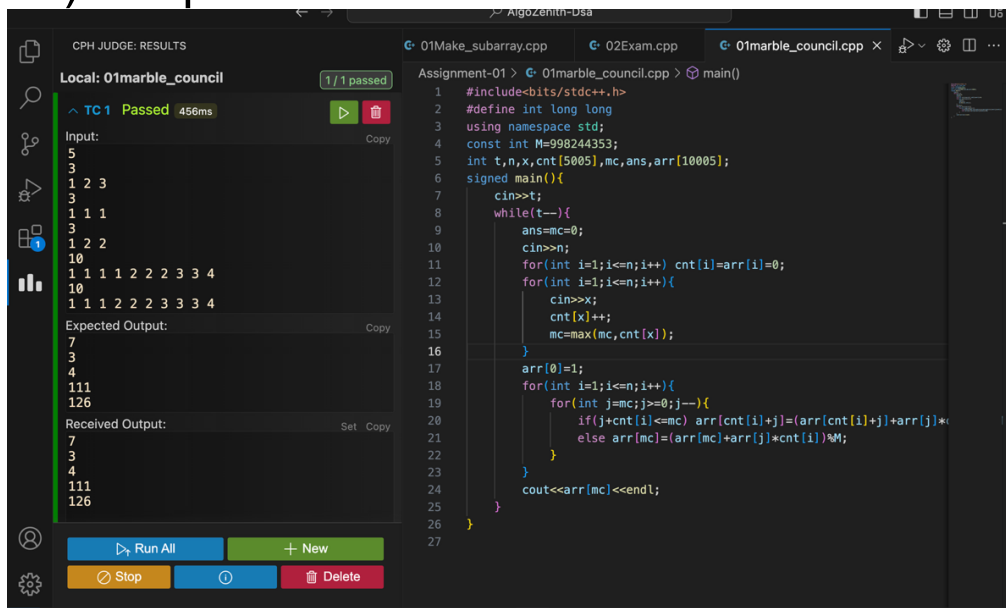
Competitive Coding-3 Assignments

Assignment – 01

01) Code:

```
#include<bits/stdc++.h>
#define int long long
using namespace std;
const int M=998244353;
int t,n,x,cnt[5005],mc,ans,arr[10005];
signed main(){
    cin>>t;
    while(t--){
        ans=mc=0;
        cin>>n;
        for(int i=1;i<=n;i++) cnt[i]=arr[i]=0;
        for(int i=1;i<=n;i++){
            cin>>x;
            cnt[x]++;
            mc=max(mc,cnt[x]);
        }
        arr[0]=1;
        for(int i=1;i<=n;i++){
            for(int j=mc;j>=0;j--){
                if(j+cnt[i]<=mc) arr[cnt[i]+j]=(arr[cnt[i]+j]+arr[j]*cnt[i])%M;
                else arr[mc]=(arr[mc]+arr[j]*cnt[i])%M;
            }
        }
        cout<<arr[mc]<<endl;
    }
}
```

01) Output:



The screenshot displays a C++ IDE with the code from the previous block and its execution results. The left sidebar shows the 'CPH JUDGE: RESULTS' for 'Local: 01marble_council', indicating '1/1 passed' with a time of 456ms. The 'Input:' section shows a sequence of numbers: 5, 3, 1 2 3, 3, 1 1 1, 3, 1 2 2, 10, 1 1 1 1 2 2 2 3 3 4, 10, 1 1 1 2 2 2 3 3 4. The 'Expected Output:' section shows: 7, 3, 4, 111, 126. The 'Received Output:' section shows the same values: 7, 3, 4, 111, 126. The main editor shows the code with line numbers 1 through 27. The code is a C++ program that processes multiple test cases, counting the frequency of numbers and calculating a result based on a combinatorial formula involving modular arithmetic.

02) Code:

```
#include<bits/stdc++.h>
using namespace std;
int t,n,m,i,arr[200005];
string s;
map<int,int>mp;
int main(){
    cin>>t;
    while(t--){
        cin>>n>>s;
        s=" "+s;
        for(i=1;i<=n;i++)arr[i]=arr[i-1]+(s[i]=='a')-(s[i]=='b');
        mp.clear();
        mp[0]=0;
        m=INT_MAX;
        for(i=1;i<=n;i++){
            mp[arr[i]]=i;
            if(mp.count(arr[i]-arr[n]))m=min(m,i-mp[arr[i]-arr[n]]);
        }
        if(m==n)m=-1;
        cout<<m<<'\n';
    }
}
```

02) Output:

The screenshot displays a C++ IDE with the following components:

- Left Panel (CPH JUDGE: RESULTS):**
 - Local: 02monocarps_string (1/1 passed)
 - TC 1 Passed (248ms)
 - Input:
 - 5
 - 5
 - bbbab
 - 6
 - bbaaba
 - 4
 - aaaa
 - 12
 - aabbbaabbaab
 - 5
 - aabaa
 - Expected Output:
 - 3
 - 0
 - 1
 - 2
 - 1
 - Received Output:
 - 3
 - 0
 - 1
 - 2
 - 1
- Right Panel (Code Editor):**
 - File: 02monocarps_string.cpp
 - Code: The same C++ code as shown in the previous block.

Assignment – 02

01) Code:

```
#include <bits/stdc++.h>
using namespace std;

int main() {
    int t;
    cin >> t;

    while (t--) {
        int a, b, c, d;
        cin >> a >> b >> c >> d;

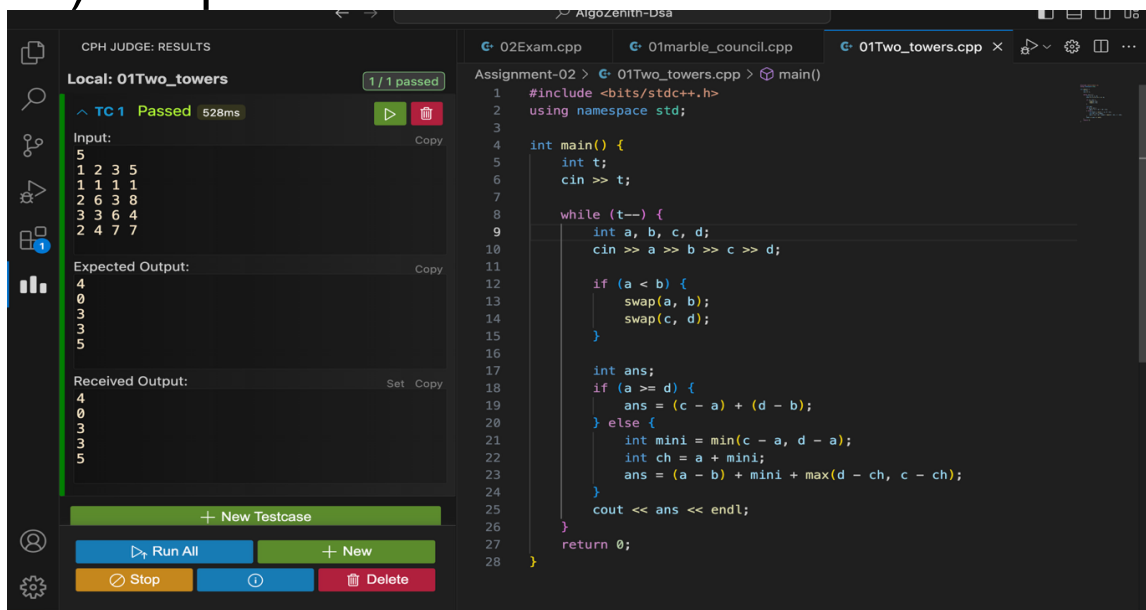
        if (a < b) {
            swap(a, b);
            swap(c, d);
        }

        int ans;
        if (a >= d) {
            ans = (c - a) + (d - b);
        } else {
            int mini = min(c - a, d - a);
            int ch = a + mini;
            ans = (a - b) + mini + max(d - ch, c - ch);
        }

        cout << ans << endl;
    }

    return 0;
}
```

01) Output:



The screenshot displays a C++ IDE with the code from the previous block on the right and its execution results on the left. The IDE has a dark theme. The left sidebar shows 'CPH JUDGE: RESULTS' for 'Local: 01Two_towers'. It indicates '1 / 1 passed' with a green checkmark and a '528ms' execution time. Below this, the 'Input:' is shown as a 5x5 grid of numbers: 5, 1 2 3 5, 1 1 1 1, 2 6 3 8, 3 3 6 4, and 2 4 7 7. The 'Expected Output:' is a 5x1 column of numbers: 4, 0, 3, 3, 5. The 'Received Output:' is identical to the expected output. At the bottom of the left panel are buttons for 'Run All', 'New', 'Stop', and 'Delete'. The right panel shows the code being executed, with line numbers 1 through 28 visible. The code is the same as in the 'Code' block.

02)Code:

```
#include <bits/stdc++.h>
using namespace std;

int main() {
    ios::sync_with_stdio(false);
    cin.tie(NULL);

    string startGene, endGene;
    cin >> startGene >> endGene;

    int n;
    cin >> n;

    vector<string> bank(n);
    for (int i = 0; i < n; i++) {
        cin >> bank[i];
    }

    unordered_set<string> st(bank.begin(), bank.end());

    if (st.find(endGene) == st.end()) {
        cout << -1 << endl;
        return 0;
    }

    queue<pair<string, int>> q;
    q.push({startGene, 0});

    vector<char> genes = {'A', 'C', 'G', 'T'};

    while (!q.empty()) {
        auto front = q.front();
        q.pop();

        string curr = front.first;
        int steps = front.second;

        if (curr == endGene) {
            cout << steps << endl;
            return 0;
        }

        for (int i = 0; i < 8; i++) {
            char original = curr[i];

            for (char ch : genes) {
                curr[i] = ch;

                if (st.count(curr)) {
```

```

        q.push({curr, steps + 1});
        st.erase(curr);
    }

    curr[i] = original;
}

cout << -1 << endl;
return 0;
}

```

02) Output:

The screenshot displays a C++ IDE with two main panels. The left panel shows the 'CPH JUDGE: RESULTS' for the 'Local: 02Min_Gen_Mutation' program, indicating that two test cases (TC 1 and TC 2) have passed. The right panel shows the source code for '02Min_Gen_Mutation.cpp'.

Test Results:

- TC 1 Passed (564ms):**
 - Input: AACCGGTT, AACCGGTA, 1, AACCGGTA
 - Expected Output: 1
 - Received Output: 1
- TC 2 Passed (179ms):**
 - Input: AACCGGTT, AACCGGTA, 3, AACCGGTA, AACCGGTA, AACCGGTA
 - Expected Output: 2
 - Received Output: 2

Source Code (02Min_Gen_Mutation.cpp):

```

1  #include <bits/stdc++.h>
2  using namespace std;
3
4  int main() {
5      ios::sync_with_stdio(false);
6      cin.tie(NULL);
7
8      string startGene, endGene;
9      cin >> startGene >> endGene;
10
11     int n;
12     cin >> n;
13
14     vector<string> bank(n);
15     for (int i = 0; i < n; i++) {
16         cin >> bank[i];
17     }
18
19     unordered_set<string> st(bank.begin(), bank.end());
20
21     if (st.find(endGene) == st.end()) {
22         cout << -1 << endl;
23         return 0;
24     }
25
26     queue<pair<string, int>> q;
27     q.push({startGene, 0});
28
29     vector<char> genes = {'A', 'C', 'G', 'T'};
30

```

Assignment – 03

01) Code:

```
#include<bits/stdc++.h>
using namespace std;
int T,n,k;
int main(){
    cin>>T;
    while(T--){
        cin>>n>>k;
        int sum=0;
        for(int i=1;i<=n;i++){
            int x;
            cin>>x;
            sum+=x;
        }
        if(sum%2==1 || (k*n)%2==0) cout<<"YES"<<endl;
        else cout<<"NO"<<endl;
    }
    return 0;
}
```

01)Output:

The screenshot displays a C++ IDE with the code from the previous block and its execution results. The left sidebar shows the 'CPH JUDGE: RESULTS' for 'Local: 01The_Equalizer', indicating '1/1 passed' with a 'TC 1 Passed' status and a time of '550ms'. The 'Input:' section lists the test cases: 4, 1 1, 1, 2 67, 67 67, 3 2, 3 3 3, 11 3, and 1 2 3 4 5 6 7 8 9 10 11. The 'Expected Output:' and 'Received Output:' sections both show 'YES YES YES NO YES YES YES NO'. The right pane shows the code for '01The_Equalizer.cpp' with line numbers 1 through 18. The code is identical to the one in the first block.

02) Code:

```
#include <bits/stdc++.h>
using namespace std;

struct Node {
    int next[26];
    int link;
    vector<int> out;
    Node() {
        memset(next, -1, sizeof(next));
        link = -1;
    }
};

vector<Node> trie(1);

int main() {
    ios::sync_with_stdio(false);
    cin.tie(NULL);

    int n;
    cin >> n;

    vector<string> s(n);
    for (int i = 0; i < n; i++) cin >> s[i];

    for (int i = 0; i < n; i++) {
        int node = 0;
        for (char c : s[i]) {
            int x = c - 'a';
            if (trie[node].next[x] == -1) {
                trie[node].next[x] = trie.size();
                trie.emplace_back();
            }
            node = trie[node].next[x];
        }
        trie[node].out.push_back(i);
    }

    queue<int> q;
    trie[0].link = 0;

    for (int c = 0; c < 26; c++) {
        if (trie[0].next[c] != -1) {
            trie[trie[0].next[c]].link = 0;
            q.push(trie[0].next[c]);
        } else {
            trie[0].next[c] = 0;
        }
    }
}
```

```

}

while (!q.empty()) {
    int v = q.front(); q.pop();
    for (int c = 0; c < 26; c++) {
        if (trie[v].next[c] != -1) {
            trie[trie[v].next[c]].link =
                trie[trie[v].link].next[c];
            q.push(trie[v].next[c]);
        } else {
            trie[v].next[c] =
                trie[trie[v].link].next[c];
        }
    }
}

long long ans = 0;

for (int i = 0; i < n; i++) {
    int node = 0;
    unordered_set<int> found;

    for (char c : s[i]) {
        node = trie[node].next[c - 'a'];

        int temp = node;
        while (temp != 0) {
            for (int id : trie[temp].out) {
                found.insert(id);
            }
            temp = trie[temp].link;
        }
    }

    found.erase(i);

    vector<int> candidates(found.begin(), found.end());

    int cnt = 0;

    for (int j = 0; j < candidates.size(); j++) {
        bool ok = true;
        for (int k = 0; k < candidates.size(); k++) {
            if (j == k) continue;

            if (s[candidates[k]].find(s[candidates[j]]) != string::npos) {
                ok = false;
                break;
            }
        }
    }
}

```



```

    }
    if (ok) cnt++;
}

ans += cnt;
}

cout << ans << endl;
}

```

02) Output:

The screenshot shows a C++ IDE with a test runner on the left and a code editor on the right.

Test Results (Left Panel):

- Local: 02Exam** (2/2 passed)
- TC 1 Passed** (563ms)
 - Input: 5, hidan, dan, hanabi, bi, nabi
 - Expected Output: 3
 - Received Output: 3
- TC 2 Passed** (184ms)
 - Input: 4, abacaba, abaca, acaba, aca
 - Expected Output: 4
 - Received Output: 4

Code Editor (Right Panel):

```

1  #include <bits/stdc++.h>
2  using namespace std;
3
4  struct Node {
5      int next[26];
6      int link;
7      vector<int> out;
8      Node() {
9          memset(next, -1, sizeof(next));
10         link = -1;
11     }
12 };
13
14 vector<Node> trie(1);
15
16 int main() {
17     ios::sync_with_stdio(false);
18     cin.tie(NULL);
19
20     int n;
21     cin >> n;
22
23     vector<string> s(n);
24     for (int i = 0; i < n; i++) cin >> s[i];
25
26     for (int i = 0; i < n; i++) {
27         int node = 0;
28         for (char c : s[i]) {
29             int x = c - 'a';
30             if (trie[node].next[x] == -1) {
31                 trie[node].next[x] = trie.size();
32                 trie.emplace_back();
33             }
34             node = trie[node].next[x];
35         }
36         trie[node].out.push_back(i);
37     }
38 }

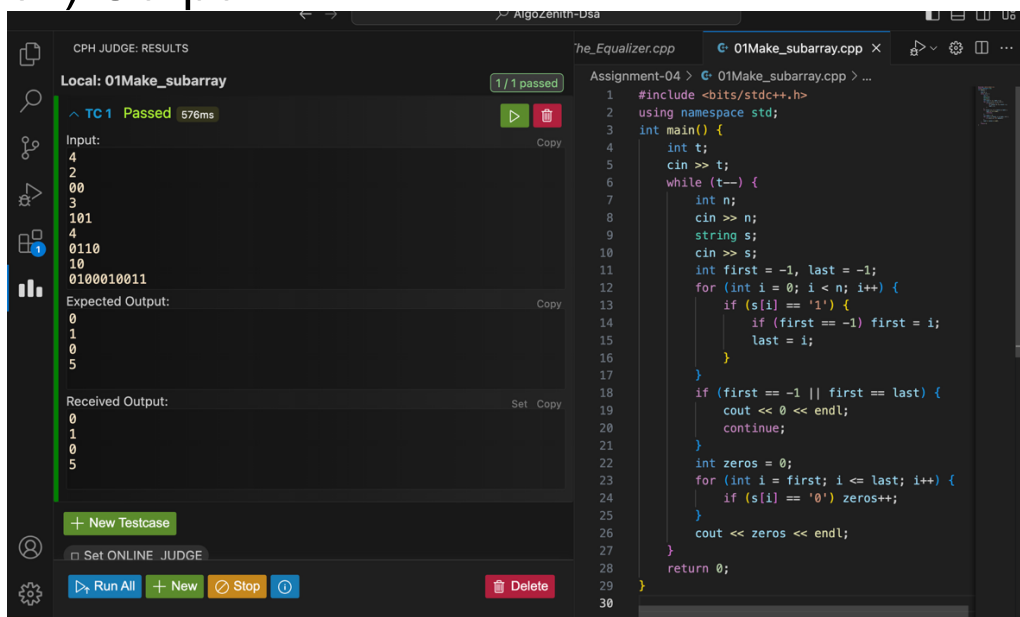
```

Assignment-04

01) Code:

```
#include <bits/stdc++.h>
using namespace std;
int main() {
    int t;
    cin >> t;
    while (t--) {
        int n;
        cin >> n;
        string s;
        cin >> s;
        int first = -1, last = -1;
        for (int i = 0; i < n; i++) {
            if (s[i] == '1') {
                if (first == -1) first = i;
                last = i;
            }
        }
        if (first == -1 || first == last) {
            cout << 0 << endl;
            continue;
        }
        int zeros = 0;
        for (int i = first; i <= last; i++) {
            if (s[i] == '0') zeros++;
        }
        cout << zeros << endl;
    }
    return 0;
}
```

01) Output:



The screenshot displays a C++ IDE with two main panels. The left panel, titled 'CPH JUDGE: RESULTS', shows the execution results for a program named '01Make_subarray'. It indicates that the program 'Passed' with a time of 576ms. The input provided is a sequence of binary strings: 4, 2, 00, 3, 101, 4, 0110, 10, and 0100010011. The expected output is 0, 1, 0, 5, and the received output matches this exactly. The right panel shows the source code of the program, which is identical to the code provided in the 'Code' section. The code is a C++ program that takes the number of test cases 't' and then processes each test case. For each test case, it reads the length of the string 'n' and the string 's' itself. It then finds the first and last positions of '1' in the string. If there are no '1's, it outputs 0. Otherwise, it counts the number of '0's between the first and last '1' (inclusive) and outputs that count.

Assignment-05

1) Code:

```
#include <iostream>
#include <queue>
#include <vector>
using namespace std;
int main() {
    int n;
    cin >> n;
    priority_queue<int, vector<int>, greater<int>> pq;
    vector<string> output;
    while (n--) {
        string s;
        int x;
        cin >> s;
        if (s == "insert") {
            cin >> x;
            pq.push(x);
            output.push_back("insert " + to_string(x));
        } else if (s == "getMin") {
            cin >> x;
            while (!pq.empty() && pq.top() < x) {
                output.push_back("removeMin");
                pq.pop();
            }
            if (pq.empty() || pq.top() > x) {
                output.push_back("insert " + to_string(x));
                pq.push(x);
            }
            output.push_back("getMin " + to_string(x));
        } else if (s == "removeMin") {
            if (pq.empty()) {
                output.push_back("insert 0");
            } else {
                pq.pop();
            }
            output.push_back("removeMin");
        }
    }
    cout << output.size() << endl;
    for (int i=0; i<output.size(); i++) {
        cout << output[i] << endl;
    }
    return 0;
}
```

1) Output:

CPH JUDGE: RESULTS

Local: Assignment_05

1 / 2 passed

TC 1 Passed 569ms

Input: 2
insert 3
getMin 4

Expected Output: 4
insert 3
removeMin
insert 4
getMin 4

Received Output: 4
insert 3
removeMin
insert 4
getMin 4

TC 2

Input: 4
insert 1
insert 1
removeMin
getMin 2

Run All + New Stop Delete

```
Assignment-05 > G Assignment_05.cpp > main()
1 #include <iostream>
2 #include <queue>
3 #include <vector>
4 using namespace std;
5 int main() {
6     int n;
7     cin >> n;
8     priority_queue<int, vector<int>, greater<int>> pq;
9     vector<string> output;
10    while (n--) {
11        string s;
12        int x;
13        cin >> s;
14        if (s == "insert") {
15            cin >> x;
16            pq.push(x);
17            output.push_back("insert " + to_string(x));
18        } else if (s == "getMin") {
19            cin >> x;
20            while (!pq.empty() && pq.top() < x) {
21                output.push_back("removeMin");
22                pq.pop();
23            } if (pq.empty() || pq.top() > x) {
24                output.push_back("insert " + to_string(x));
25                pq.push(x);
26            } output.push_back("getMin " + to_string(x));
27        } else if (s == "removeMin") {
28            if (pq.empty()) {
29                output.push_back("insert 0");
30            } else {
```

CPH JUDGE: RESULTS

insert 4
getMin 4

TC 2 Passed 407ms

Input: 4
insert 1
insert 1
removeMin
getMin 2

Expected Output: 6
insert 1
insert 1
removeMin
removeMin
insert 2
getMin 2

Received Output: 6
insert 1
insert 1
removeMin
removeMin
insert 2
getMin 2

Run All + New Stop Delete

```
Assignment-05 > G Assignment_05.cpp > main()
1 #include <iostream>
2 #include <queue>
3 #include <vector>
4 using namespace std;
5 int main() {
6     int n;
7     cin >> n;
8     priority_queue<int, vector<int>, greater<int>> pq;
9     vector<string> output;
10    while (n--) {
11        string s;
12        int x;
13        cin >> s;
14        if (s == "insert") {
15            cin >> x;
16            pq.push(x);
17            output.push_back("insert " + to_string(x));
18        } else if (s == "getMin") {
19            cin >> x;
20            while (!pq.empty() && pq.top() < x) {
21                output.push_back("removeMin");
22                pq.pop();
23            } if (pq.empty() || pq.top() > x) {
24                output.push_back("insert " + to_string(x));
25                pq.push(x);
26            } output.push_back("getMin " + to_string(x));
27        } else if (s == "removeMin") {
28            if (pq.empty()) {
29                output.push_back("insert 0");
30            } else {
```